Subject outline

Lecture topics (by week)

1 – Introduction to quantum computing and maths basics
2 – Single qubit representations and logic operations
3 – Two qubit states and logic gates
4 – Multi-qubit states and quantum arithmetic
5 – Basic quantum algorithms
6 – Period finding, cryptography and quantum factoring
7 – Shor's algorithm, post-quantum crypto, quantum key distribution
8 – Quantum search algorithms
9 – Grover search applications, optimisation problems
10 – Solving optimisation problems on quantum computers
11 – Applications in quantum machine learning
12 – Real quantum computer devices

Assignment schedule:
#1: Hand out in Week 2
#2: Hand out in Week 8

Week 6

Lecture 11
11.1 Introduction to Cryptography
11.2 Public Key Cryptography
11.3 RSA Cryptography
11.4 Security of RSA
11.5 Factoring and Period Finding

Lecture 12
12.1 Quantum Phase Estimation
12.2 Quantum Fourier Transform
12.3 Phase as a periodic function

Practice class 6
RSA and Quantum Phase Estimation

Recap: Story of Cryptography

Alice (Sender)

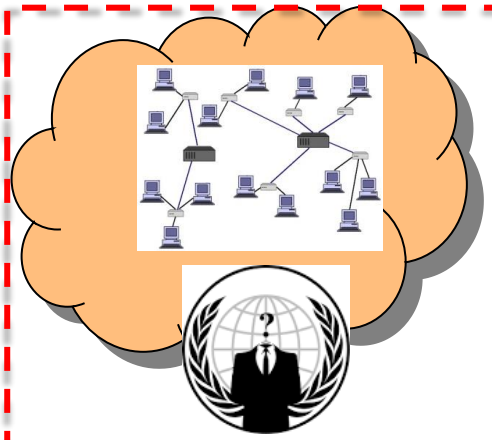
- Uses an Encryption Function (E)

m (Plain Text) → E

Ke → E

Encryption Key (Public/Private)

C = Sent ciphertext



Eve (Adversary)

- Uses network probes to attack communication

Tampering & eavesdropping

C' = Received Cipher Text (Could be in error)

Bob (Receiver)

- Uses a Decryption Function (D)

Kd → D: Decryption Key (Private);

D → m' (Recovered Plain Text)

E, D are public; c is the ciphertext, c' is received ciphertext; ideally m=m';

Cryptography involves many conceptual ideas, we look at the basic functions

Images: General Internet resources

Recap: Factoring and Period finding

The security of RSA comes down to factoring. In turn factoring can be reduced to period finding:

If we can find the period of

$$f(k) = a^k \bmod N$$

efficiently, we can factor efficiently.

Classically, we do not have any efficient algorithm for the period finding problem. This lecture we'll look at period finding on a quantum computer, extending what we've already seen in Simon's algorithm.

12.1 Quantum Phase Estimation

The Problem

Consider an eigenvector $|\psi\rangle$ of a unitary matrix (an operation which you could implement on a quantum computer) U:

$$\boxed{U\,|\psi\rangle = \underbrace{\exp(2\pi i\theta)}_{\text{Eigenvalue}}\,\underbrace{|\psi\rangle}_{\text{Eigenvector}}}$$

The Quantum Phase Estimation algorithm estimates the angle θ. Notice that since U is unitary, all eigenvalues of U will be of this form.

The T gate

For example, consider the T gate:

$$T = \begin{bmatrix} 1 & 0 \\ 0 & \exp(i\frac{\pi}{4}) \end{bmatrix}$$

An eigenvector of the T gate is

$$\boxed{T\,|1\rangle = \underbrace{\exp\left(i\frac{\pi}{4}\right)}_{\text{Eigenvalue}}\,\underbrace{|1\rangle}_{\text{Eigenvector}}}$$

$$\boxed{U\,|\psi\rangle = \exp(2\pi i\theta)\,|\psi\rangle}$$

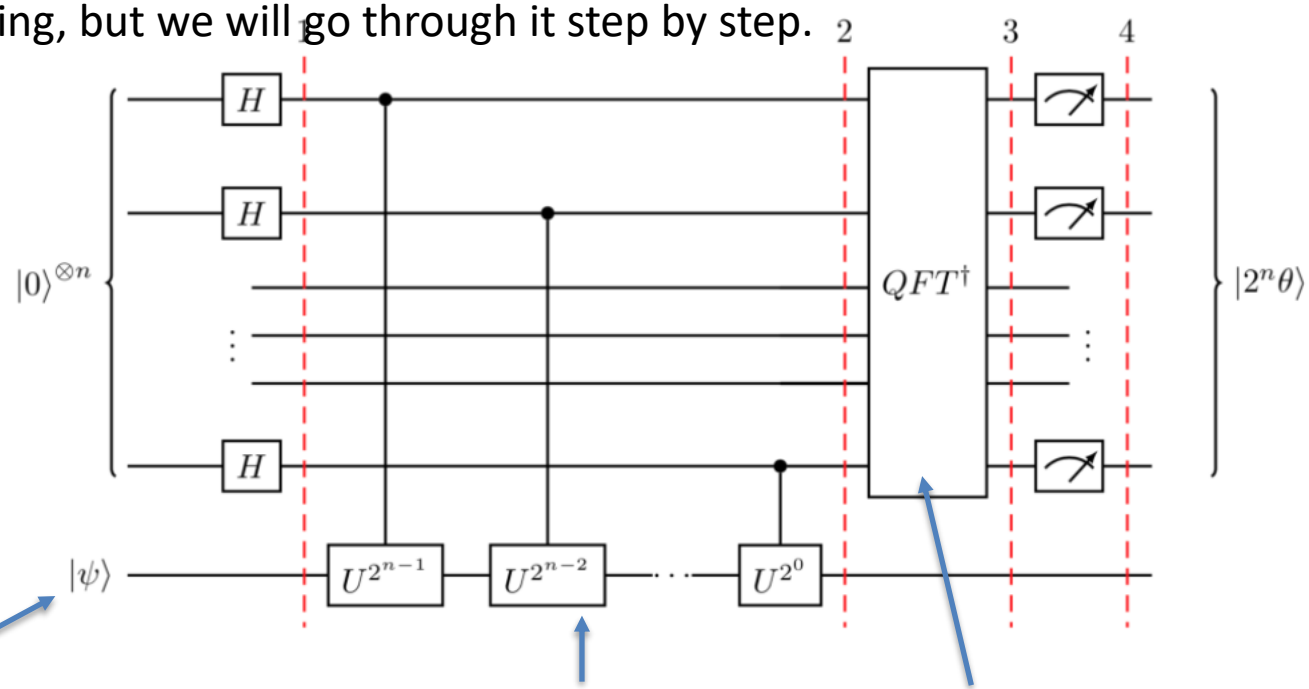
So in this case, we want to find:

$$\theta = \frac{1}{8}$$

Quantum Phase Estimation gives a way to do this for arbitrary U on a quantum computer.

Quantum Phase Estimation Circuit

Looks daunting, but we will go through it step by step.



Measure an estimate of θ

Eigenstate

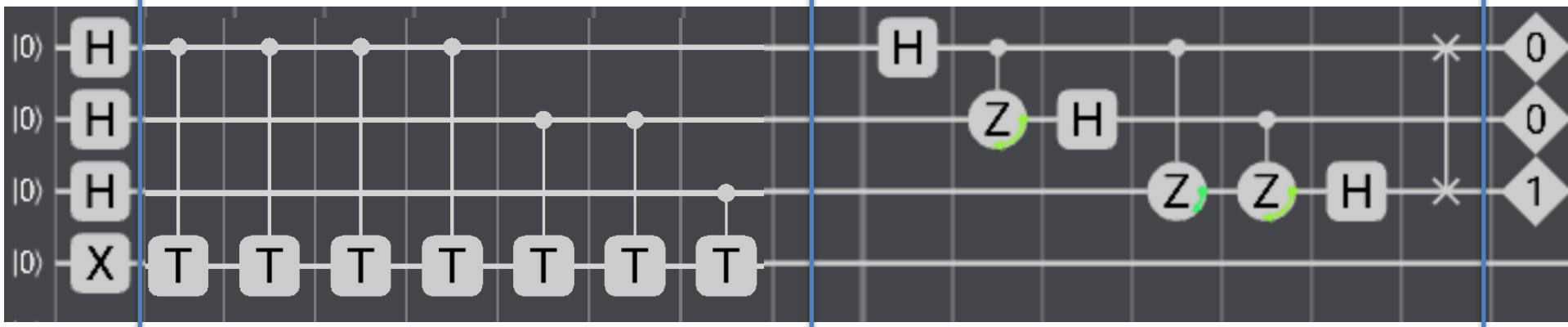
Operation, U (controlled on upper register)

Inverse QFT

(1) Prepare equal superposition, x in the upper register x and eigenstate $|\psi\rangle$.
(2) Apply U^x to lower register
(3) Apply (inverse) Quantum Fourier Transform (we will treat this as a black box) to the upper register
(4) Measure to obtain an estimate equal to $2^n\theta$.

QPE Circuit in the QUI

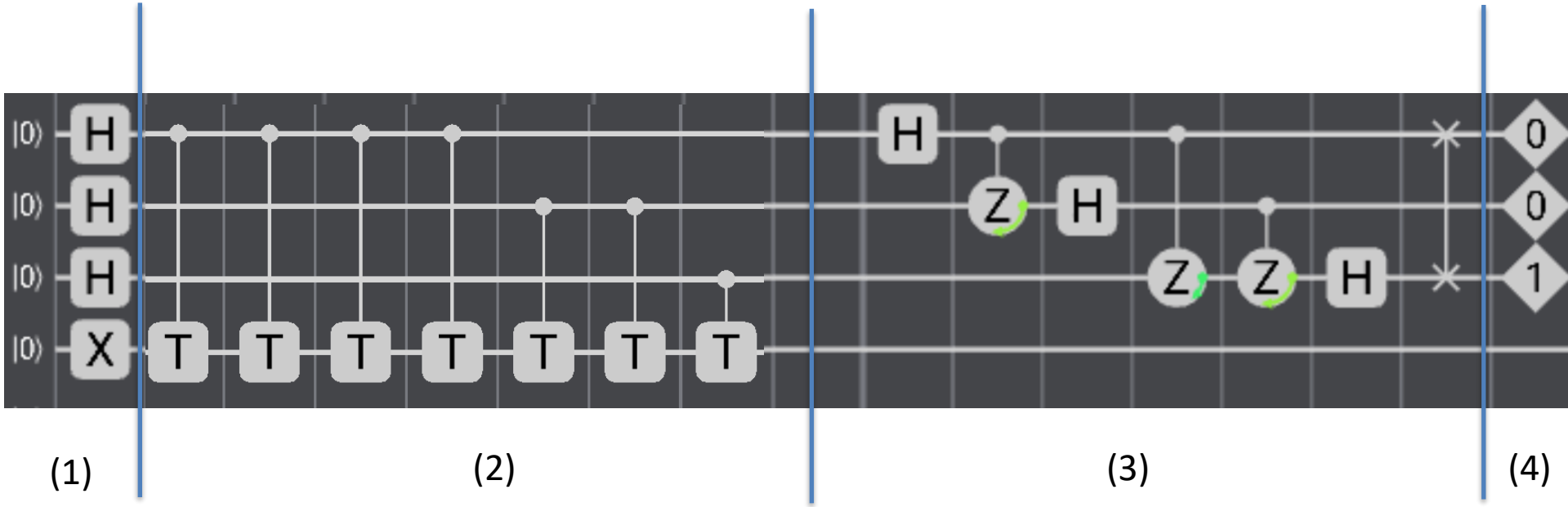
$$T\,|1\rangle = \exp\left(i\frac{\pi}{4}\right)|1\rangle$$



(1) (2) (3) (4)

(1) Prepare equal superposition, x and eigenstate $|\psi\rangle$.
(2) Apply U^x to lower register
(3) Apply (inverse) Quantum Fourier Transform to the x register
(4) Measure to obtain an estimate equal to $2^n\theta$.

Step 1: Equal Superposition

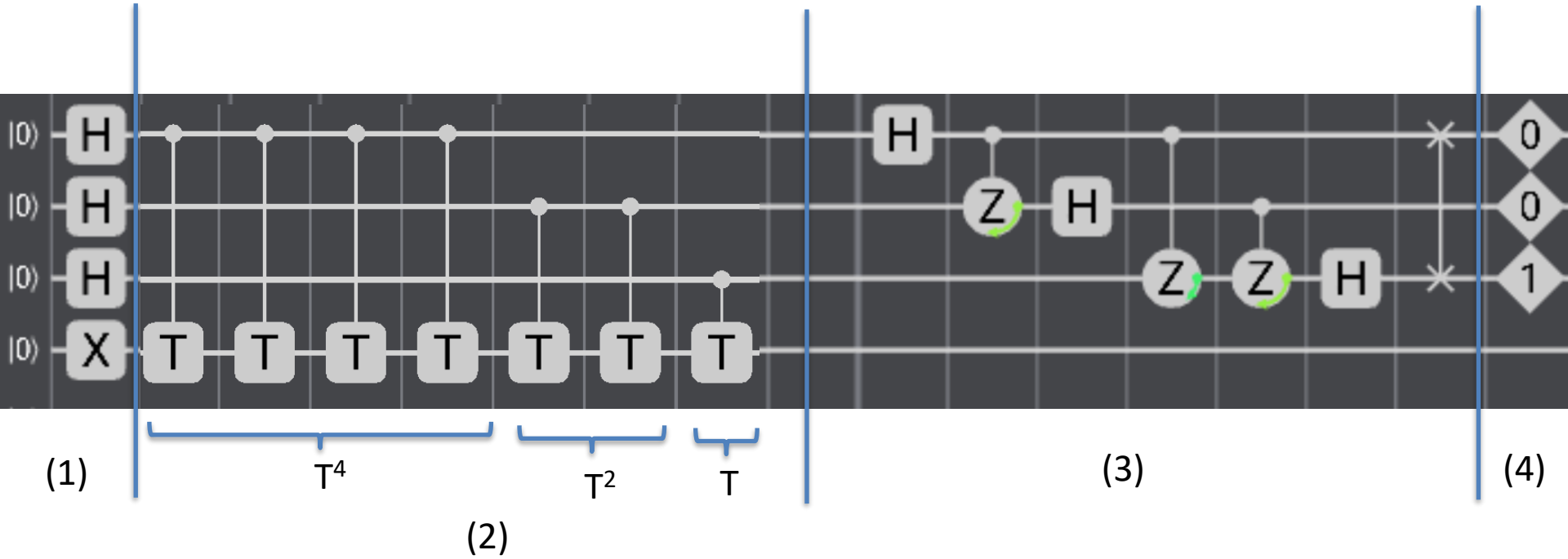


After the first step of the algorithm

$$|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle\,|1\rangle$$

(x is in decimal notation)

Step 2: Applying T^x



After step 2:

$$|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle\, T^x\, |1\rangle$$

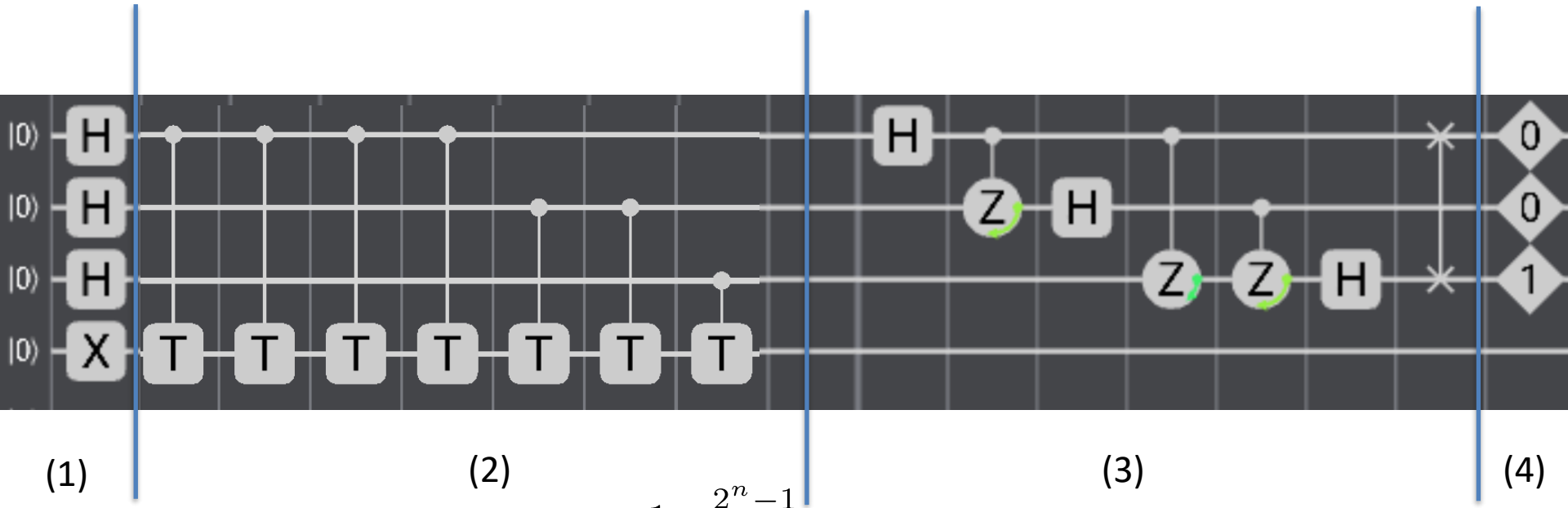
$$= \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \exp\left(i\frac{\pi}{4}x\right) |1\rangle$$

$$T\,|1\rangle = \exp\left(i\frac{\pi}{4}\right)|1\rangle$$

$$T^2\,|1\rangle = \exp\left(i\frac{\pi}{2}\right)|1\rangle$$

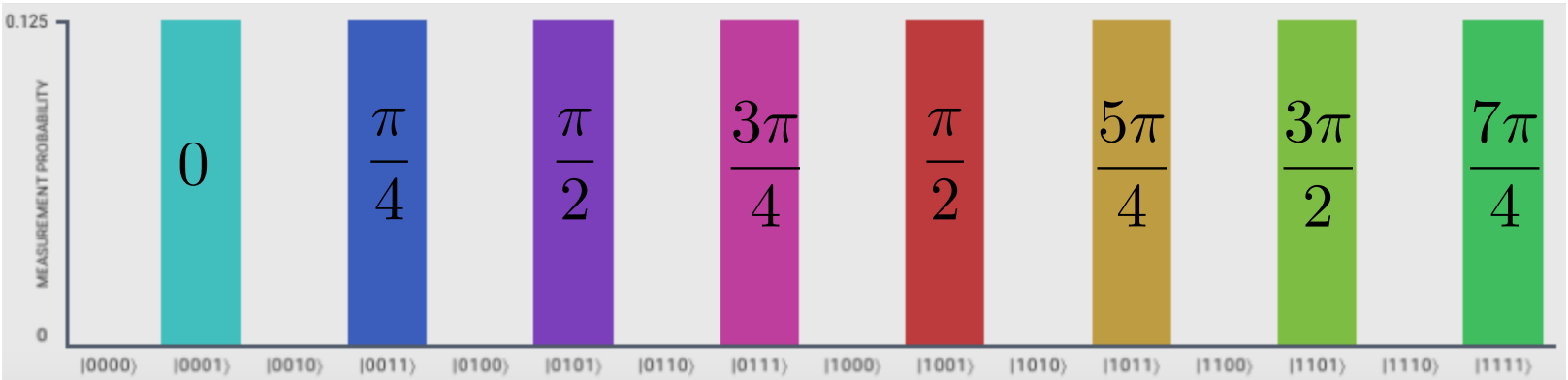
ect.

Step 2: Applying T^x



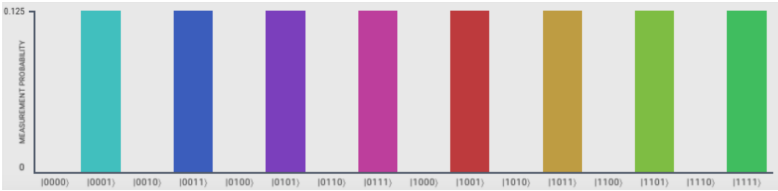
$$|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} \exp\left(i\frac{\pi}{4}x\right) |x\rangle |1\rangle$$

Equal superposition, where each state has a phase proportional to x:

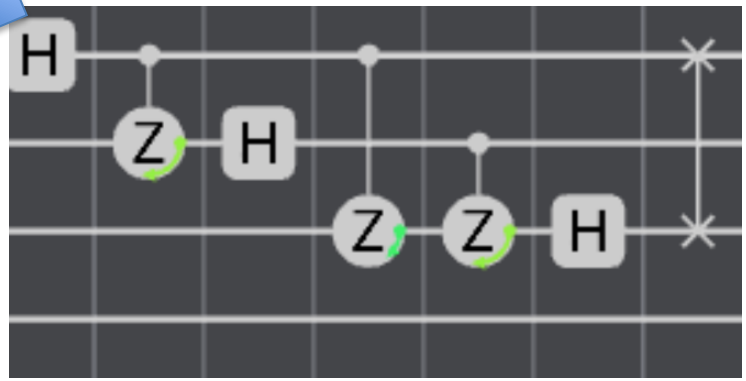


The Fourier Transform

Original state



Quantum Fourier Transform

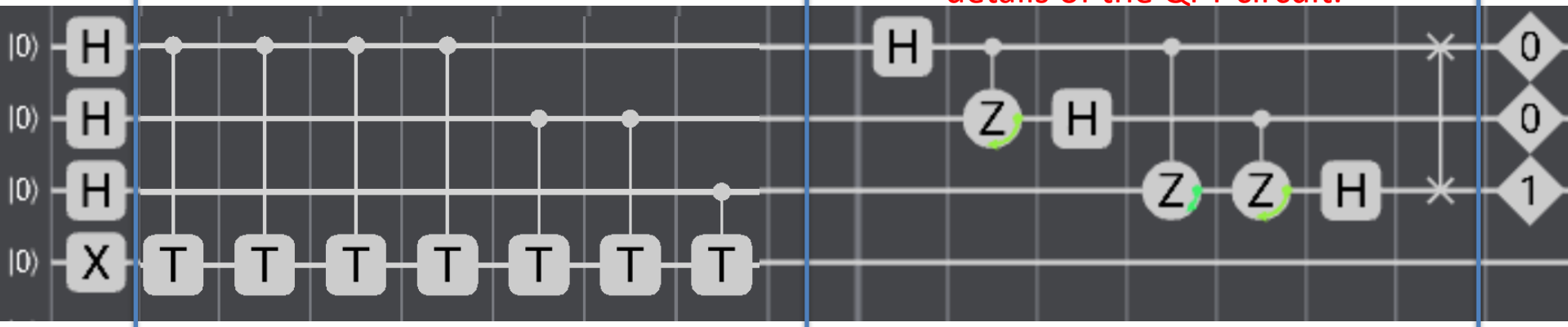


Resulting state

$$2^3\theta = 1$$

Step 3: Quantum Fourier Transform

You will not be tested on the details of the QFT circuit.



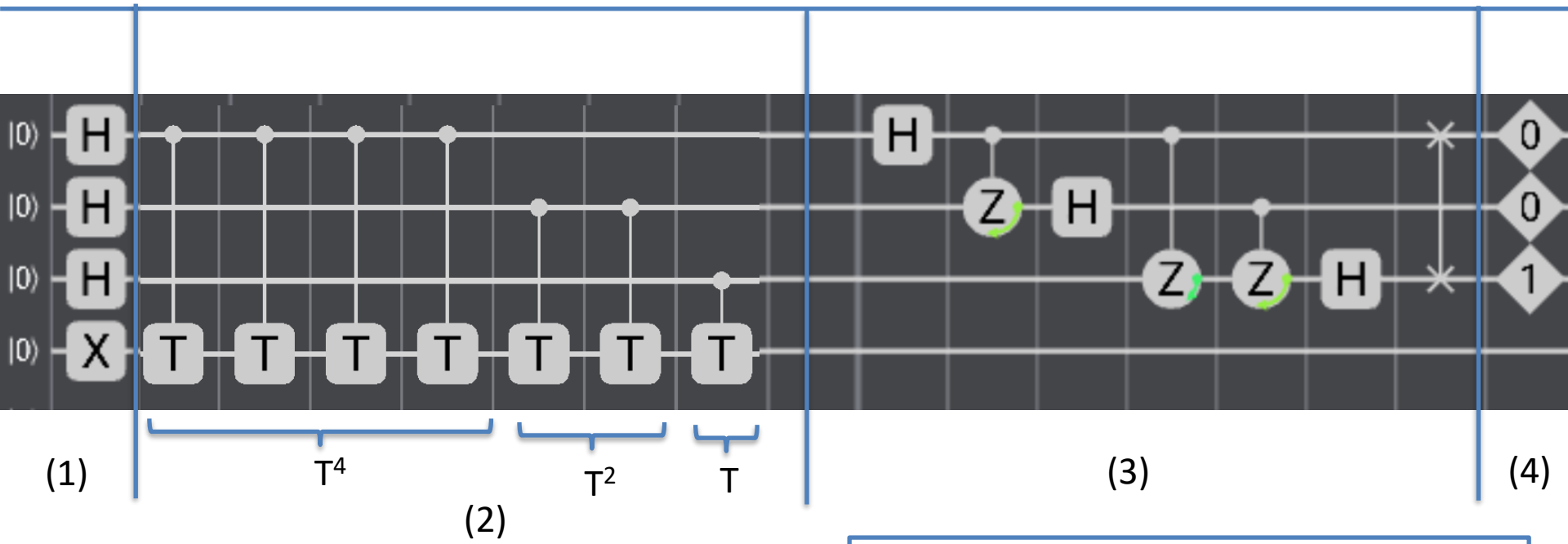
(1) (2) (3) (4)

$$|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} \exp\left(i\frac{\pi}{4}x\right) |x\rangle |1\rangle$$

The QFT determines the period of the function, in this case it will exactly find the answer:

$$|\psi\rangle = |001\rangle |1\rangle$$

Step 4: Measure



Finally we measure an obtain the result 001_2=1, so in this case:

$$2^n \theta = 1$$

$$\theta = \frac{1}{8}$$

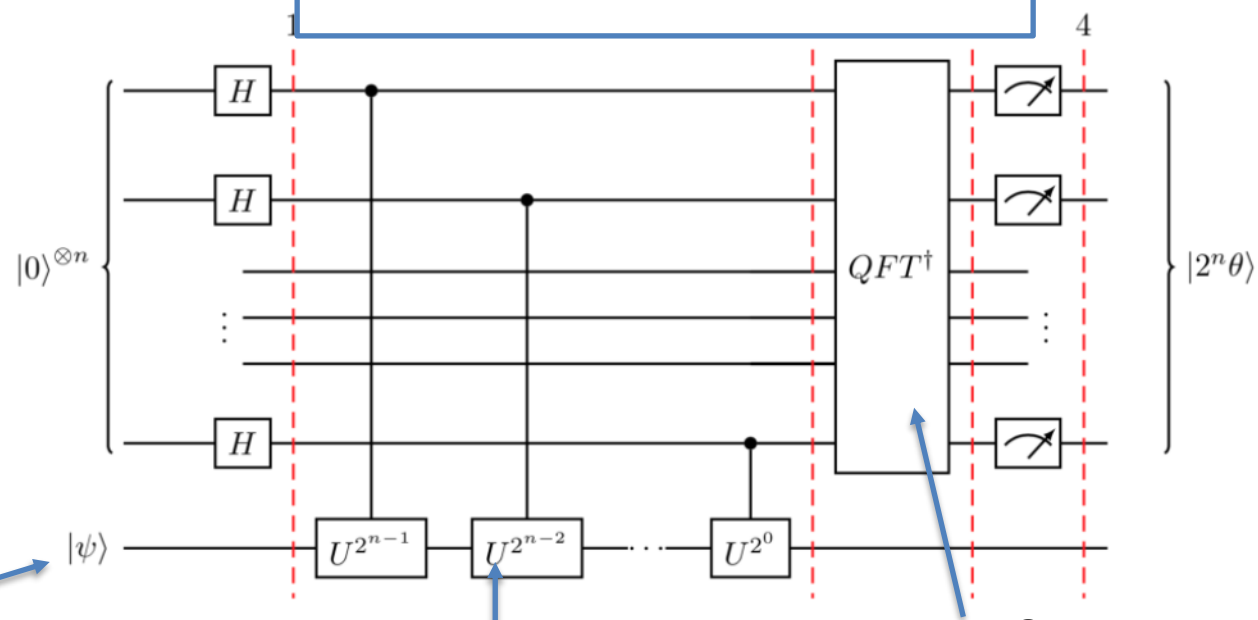
As we expected!

$$U\,|\psi\rangle = \exp(2\pi i\theta)\,|\psi\rangle$$

$$T\,|1\rangle = \exp\left(i\frac{\pi}{4}\right)|1\rangle$$

Quantum Phase Estimation Circuit

$$U\,|\psi\rangle = \exp(2\pi i\theta)\,|\psi\rangle$$



Measure an estimate of θ

Eigenstate

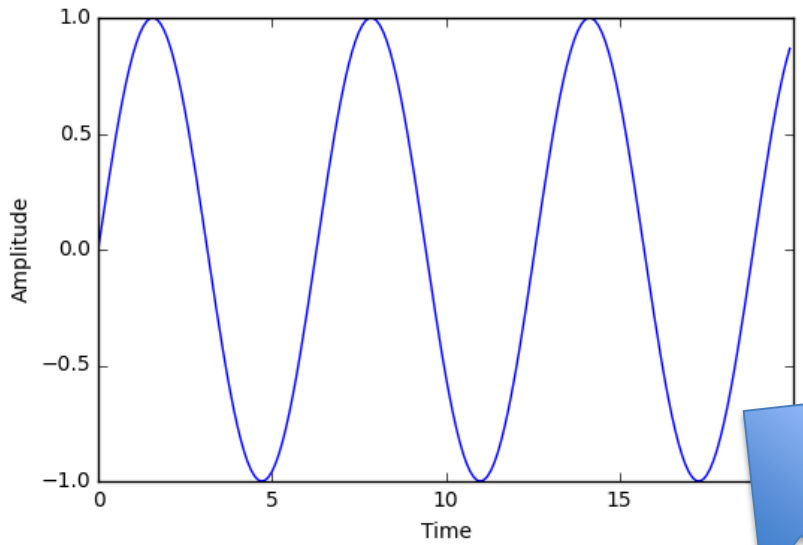
Operation, U

Inverse QFT

(1) Prepare equal superposition, x and eigenstate $|\psi\rangle$.
(2) Apply U^x to lower register
(3) Apply (inverse) Quantum Fourier Transform to the x register
(4) Measure to obtain an estimate equal to $2^n\theta$.

12.2 Quantum Fourier Transform
(will not be examined)

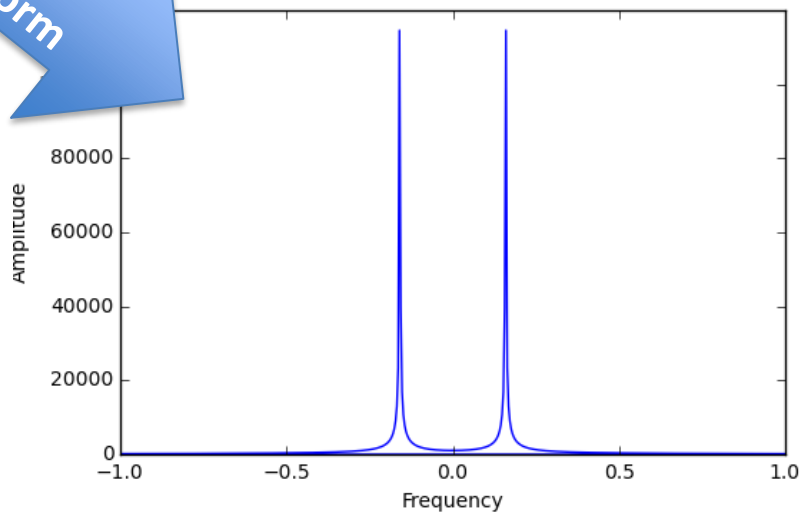
Introduction to Fourier Transform



Time domain

Fourier Transform

Frequency Domain



Discrete Fourier Transform

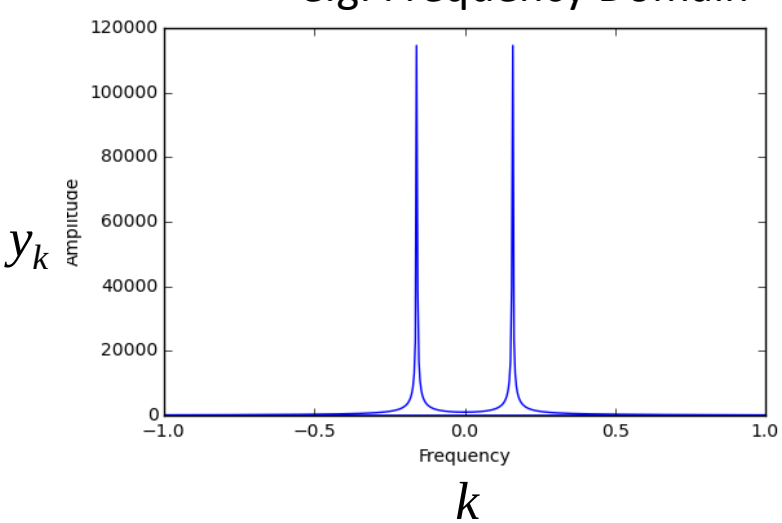
Maps a vector: $(x_0, x_1, ..., x_{N-1}) \in \mathcal{C}^N$ to a vector: $(y_0, y_1, ..., y_{N-1}) \in \mathcal{C}^N$

According to:

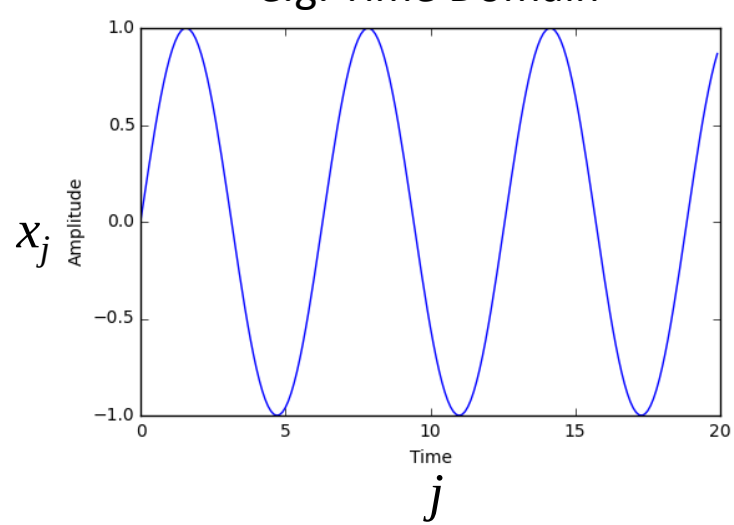
$$y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j e^{2\pi i j k / N}$$

NB.
i = sqrt(-1)
j and k are integers

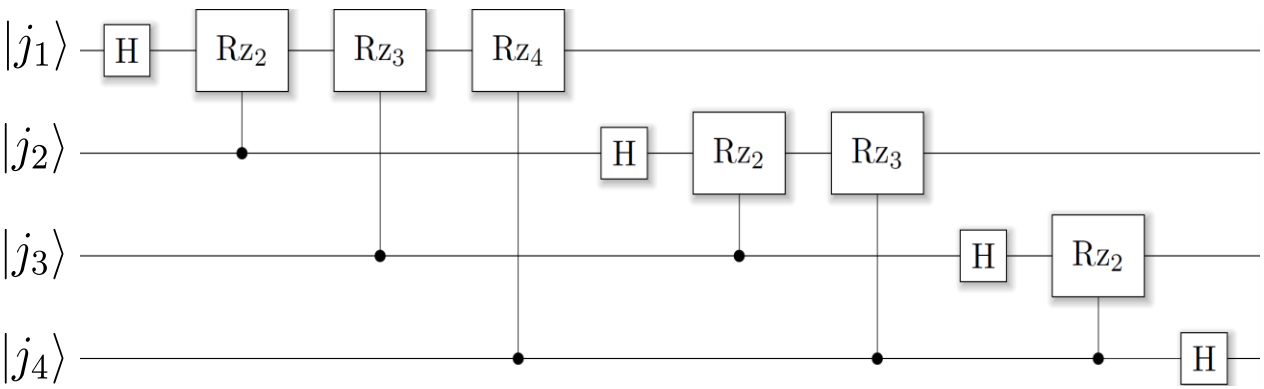
e.g. Frequency Domain



e.g. Time Domain



Circuit for QFT



$$\frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i\, 0.j_1 j_2 j_3 j_4}\,|1\rangle\right) \rightarrow |j_4\rangle$$

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$$\frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i\, 0.j_3 j_4}\,|1\rangle\right) \rightarrow |j_2\rangle$$

$$\frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i\, 0.j_4}\,|1\rangle\right) \rightarrow |j_1\rangle$$

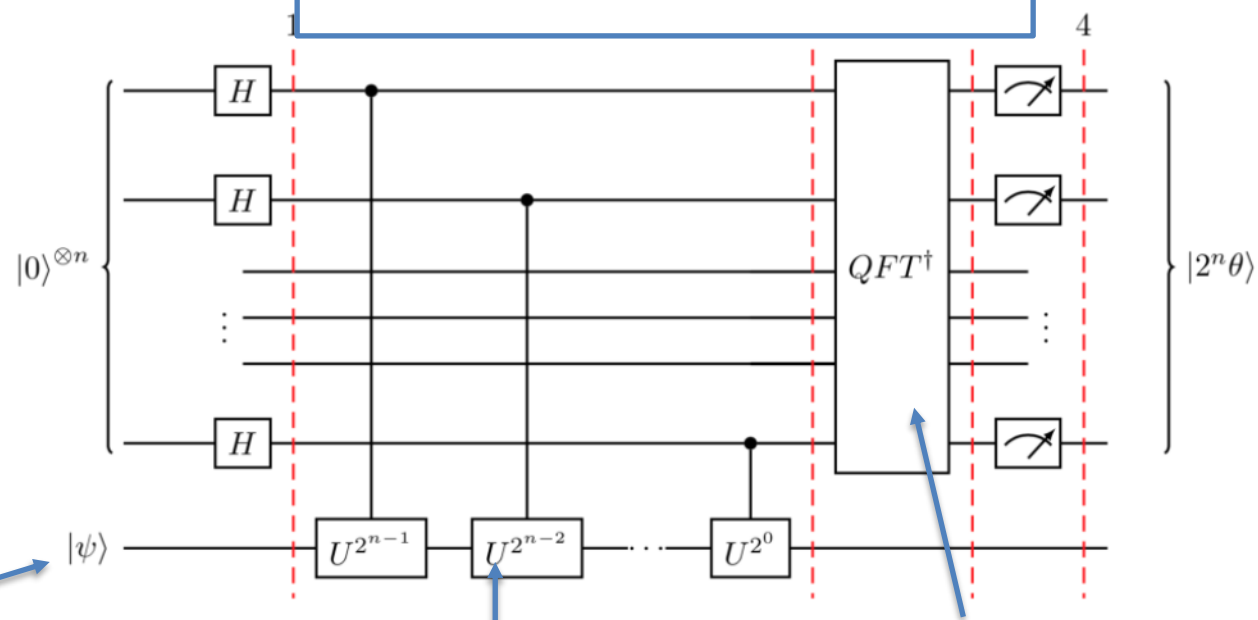
Controlled rotations with: $R_{z_k} = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{pmatrix}$

Notice how the required QFT form is recovered by re-labelling qubits

12.3 Phase as a periodic function

Quantum Phase Estimation Circuit

$$U\,|\psi\rangle = \exp(2\pi i\theta)\,|\psi\rangle$$



Measure an estimate of θ

Eigenstate

Operation, U

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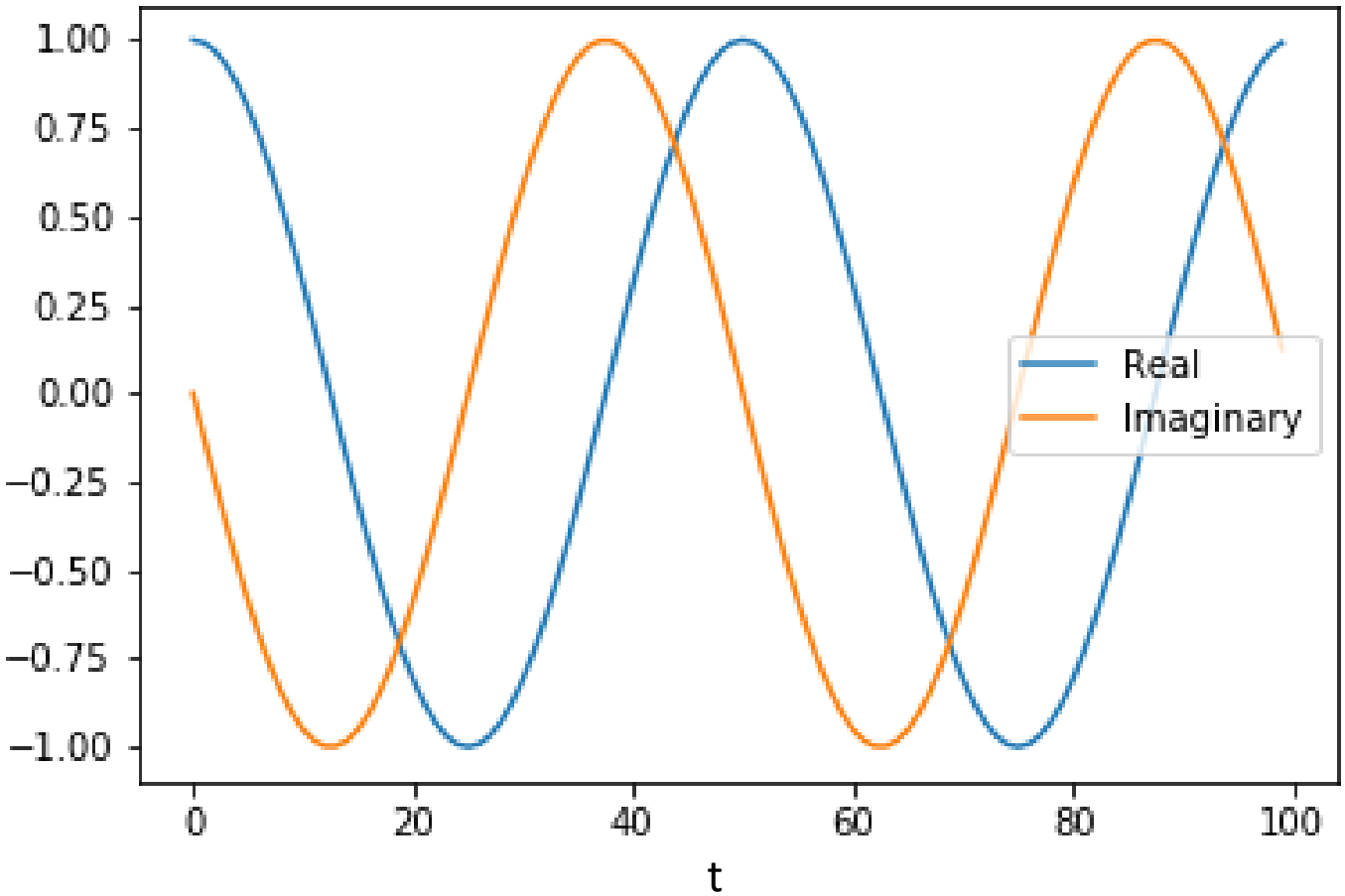
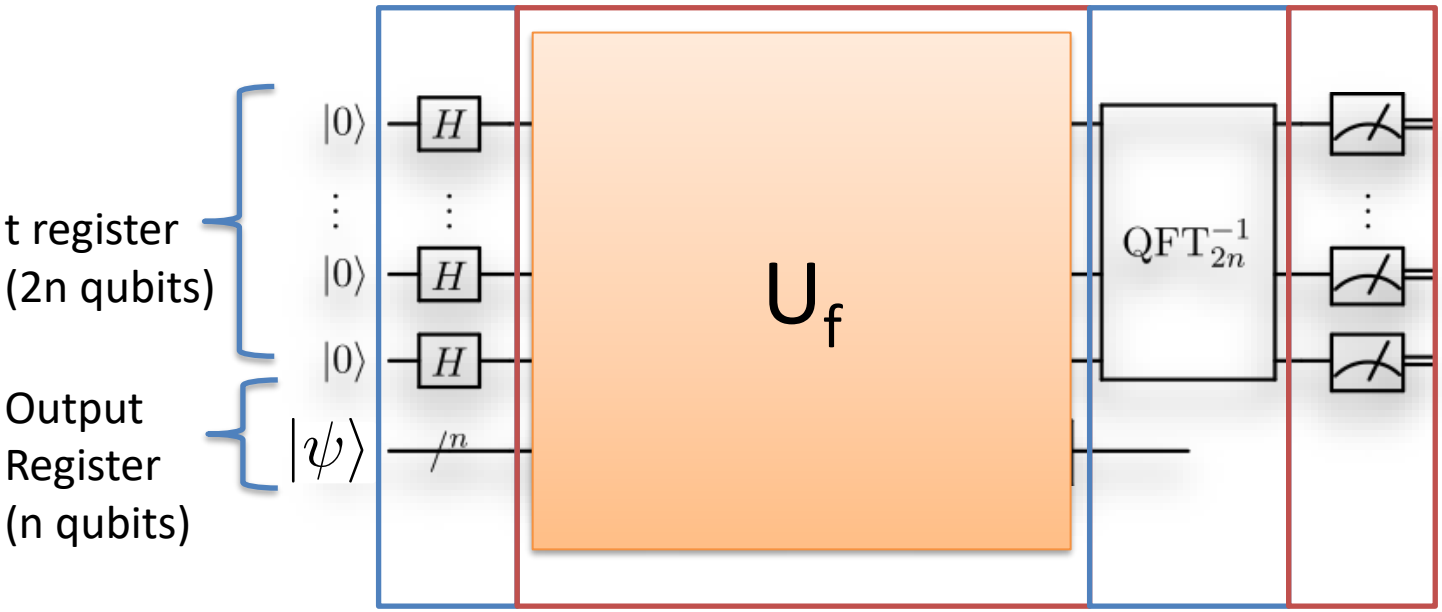
The phase is periodic

$$f(t) = \exp\left(2\pi i \theta t\right)$$



Quantum Phase Estimation as a function



$$U\,|t\rangle\,|\psi\rangle = |t\rangle\,\exp(i2\pi\theta t)\,|\psi\rangle$$

Quantum computers find periods of periodic functions efficiently!

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